

Khulna University of Engineering & Technology
Department of Computer Science and Engineering

**A Machine Learning Approach to
Scoring and Ranking University Websites**
A Seven-Dimension Scoring Model and a Comparative Study
of Twelve Regression Algorithms

Course: Machine Learning Laboratory
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Abstract

University websites are the primary interface between an institution and a prospective student, yet their quality is rarely measured systematically. This project develops an automated method for scoring and ranking them.

We collected 1,230 university landing pages worldwide, each described by 69 measurable attributes, and reduced them to a clean modelling dataset of **1,225 universities and 71 features**. We then defined a **seven-dimension scoring model** that converts those attributes into a website quality score from 0 to 100. The dimensions are organised around what a prospective student needs from the site — what can be studied, how to apply, whether the institution is active, whether the site can be navigated, used, trusted and read — and are combined as a weighted sum subject to two gating rules. Four attributes are mapped through explicit non-linear response curves, because their relationship with quality is not monotone.

The dataset was split **80% training / 20% testing**, stratified on score band, and **twelve regression algorithms** were trained and compared under identical conditions. Performance ranged from $R^2 = 0.786$ (k-nearest neighbours) to $R^2 = 0.992$ (LightGBM). After grid-search tuning, the selected LightGBM model achieves $R^2 = 0.993$, MAE = 1.11 score points and Spearman $\rho = 0.993$ on the 246 held-out universities.

The comparative study yields a finding of practical interest. The separation between algorithm families is not random: it is concentrated entirely on the universities to which a gating rule applies. On those 36 test cases, linear regression is wrong by 6.41 points on average against 1.07 for LightGBM — a 6× difference — because a weighted sum cannot express a hard cap while a decision tree can. Gain-based importance analysis confirms that the tuned model independently recovered the gate: the single navigation attribute that triggers it accounts for 26% of all loss reduction.

Keywords: website quality assessment, regression, model comparison, feature engineering, gradient boosting, university ranking.

Contents

1	Introduction	2
1.1	Motivation	2
1.2	Problem statement	2
1.3	Objectives	2
1.4	Contributions	2
2	Dataset and Preparation	2
2.1	Source data	2
2.2	Cleaning	3
2.3	Missing values	3
2.4	Feature engineering	3
3	The Scoring Model	4
3.1	Design principle	4
3.2	Gating rules	4
3.3	Non-linear response curves	5
3.4	The resulting target	5
4	Experimental Setup	6
4.1	Train / test split	6
4.2	Algorithms compared	6
4.3	Evaluation metrics	6

5	Model Comparison	6
5.1	Cross-validation	7
6	Hyper-parameter Tuning and Final Evaluation	7
7	Why the Algorithms Differ	8
8	What Drives a Website’s Score	9
9	Rankings	11
10	Limitations	11
11	Conclusion	11
A	Repository contents	12

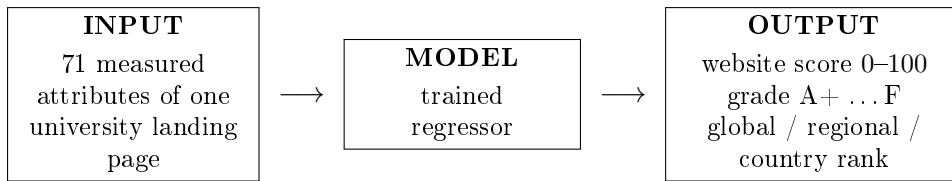
1 Introduction

1.1 Motivation

For a prospective student, a university’s website is often the only source of information available before applying. It determines whether they can find the programmes on offer, learn the admission requirements, discover a deadline, or contact a human being. A poorly organised site is not a cosmetic problem; it is a barrier to access.

Despite this, website quality is usually assessed by manual inspection — slow, subjective and impossible to apply at scale. This project asks whether the assessment can be automated: whether a set of attributes that a program can extract from a landing page is sufficient to predict a meaningful quality score.

1.2 Problem statement



This is a **supervised regression** problem. Given the attribute vector of a landing page, predict its website quality score. Because the deliverable is a league table, rank-order agreement (Spearman ρ) is treated as the primary metric alongside R^2 and mean absolute error.

1.3 Objectives

1. Collect and clean a global dataset of university landing-page attributes.
2. Define a transparent, reproducible scoring model that converts attributes into a 0–100 website quality score reflecting a prospective student’s needs.
3. Split the data 80/20 and train a broad range of regression algorithms on identical inputs.
4. Compare their performance and, more importantly, *explain* why they differ.
5. Produce global, regional and country rankings, and analyse what drives a website’s score.

1.4 Contributions

1. A seven-dimension scoring model with documented weights, two gating rules and four non-linear response curves (Section 3).
2. A controlled comparison of twelve regression algorithms across six model families on the same 80/20 split (Section 5).
3. An error analysis that attributes the performance gap between families to a specific, demonstrable mechanism rather than to model capacity in the abstract (Section 7).
4. A ranking of 1,225 university websites with an interpretable account of what drives the score (Sections 8–9).

2 Dataset and Preparation

2.1 Source data

The raw dataset consists of 1,230 university landing pages, each described by 69 attributes extracted automatically, organised into eleven groups: header and navigation, rankings and recognition, notices and updates, events and media, page content, footer, visual design, service interaction, technical performance, SEO metadata and accessibility. Attributes are of three kinds: presence

flags (does a programmes page exist), counts (how many menu items) and measurements (contrast ratio, mobile score, load time in seconds).

2.2 Cleaning

Every step below is implemented in `src/build_dataset.py` and is reproducible from the raw file. Nothing was corrected silently.

Table 1: Data preparation log.

Issue	Scale	Action taken
Constant or empty columns	3 columns	Dropped: they carry no information
Exact duplicate column	<code>a62</code> $\equiv$ <code>load_time_s</code>	One copy dropped
External ranking <i>values</i>	95–99% missing	Dropped: unusable, and they describe the institution rather than the website. The ranking <i>badge</i> flags are kept, since displaying a badge is a property of the page
Duplicate sites	10 rows $\rightarrow$ 5	Deduplicated on the <i>resolved</i> URL, so two entry points to the same page are recognised as one site
Notice dates after the crawl date	284 rows	Censored to the crawl date rather than deleted
Regional labels in the country field	169 rows	Flagged; country rankings exclude them
Final dataset	1,225 universities $\times$ 71 features	

2.3 Missing values

Three attributes are missing for some universities. The conventional remedy — replace with the column median — is inappropriate for all three, and for one of them it actively inverts the meaning of the data.

Why median imputation was rejected

The median of `notice_recency_days` is **1 day**. Median-filling would therefore record the **468 universities that have no dated notice anywhere on their site** as having posted something *yesterday* — assigning the strongest freshness signal in the dataset to precisely the institutions that earned it least. Absence would be read as *average* when it in fact means *absent*.

Because absence has a known direction for each of the three, they are filled at the worst defensible value, and a companion `*_was_missing` indicator is computed *before* the fill so that “never measured” remains recoverable as a separate fact.

Attribute	Rows	Filled with	What absence means
<code>notice_recency_days</code>	468	3650 days	no dated notice exists anywhere
<code>a72_alt_text_pct</code>	4	0%	no image carries a text alternative
<code>a53_contrast_ratio</code>	1	1.0:1	no readable text block was found

2.4 Feature engineering

Four derived features are added. `notice_recency_days` converts a notice date into a staleness measure. `load_time_z_region` standardises load time within region, because raw load time is

confounded with the data collector: each of six collectors covered exactly one region, and regional medians span 8.1 s to 12.9 s, so part of the difference is measurement rather than website. `broken_links_log` compresses a heavily skewed count. Three `*_was_missing` indicators record measurement failure.

3 The Scoring Model

3.1 Design principle

The dataset records what a website *has*; it does not record how good it is. A scoring model is therefore required to convert attributes into a score. Ours is organised around a single question:

What does a prospective student need from a university website?

Seven dimensions answer it. Each is a sub-score $D_k \in [0, 1]$ computed from named attributes, and the website score is their weighted sum subject to a cap:

$$\text{Score} = \min\left(\sum_{k=1}^7 w_k D_k, \text{cap}\right), \quad \sum_{k=1}^7 w_k = 100 \quad (1)$$

Table 2: The seven dimensions, their weights, and the attributes that compose them.

	Dimension	w_k	Principal attributes
D_1	Academic information	28	programmes listing, department links, faculty page, library, research highlight, careers
D_2	Admission support	22	admissions policy, admission notices, scholarships, contact page, prospectus
D_3	Currency and activity	15	notice recency, dated events, news section, academic calendar
D_4	Navigation and findability	15	primary navigation, menu breadth, search box, breadcrumb, quick links, sitemap
D_5	Usability and accessibility	10	mobile score, contrast ratio, alt-text coverage, accessible design, language toggle
D_6	Technical quality	7	HTTPS, broken links, load speed relative to region
D_7	Institutional transparency	3	vision/mission, about text, footer contact, accreditation, student portal
	Total	100	

The weights are ordered by how directly each dimension serves the applicant’s task. The content a student actually came for accounts for half the score ($D_1 + D_2 = 50$); the machinery that delivers it accounts for most of the remainder; self-presentation accounts for the least. The weights were fixed before any score was computed.

3.2 Gating rules

Two failures are severe enough that no amount of other content compensates:

Condition	Cap	Rationale
no primary navigation	45	A site that cannot be navigated cannot serve an applicant, however much content is buried inside it
no HTTPS	60	A site that is not encrypted should not be trusted with an application form or personal data

A gate applies to **186 of 1,225 universities (15.2%)**. Gates make Equation 1 non-linear, and Section 7 shows that this single design choice accounts for most of the difference between the algorithms compared.

3.3 Non-linear response curves

Four attributes do not reward linearly, and the scoring model states this explicitly rather than assuming proportionality.

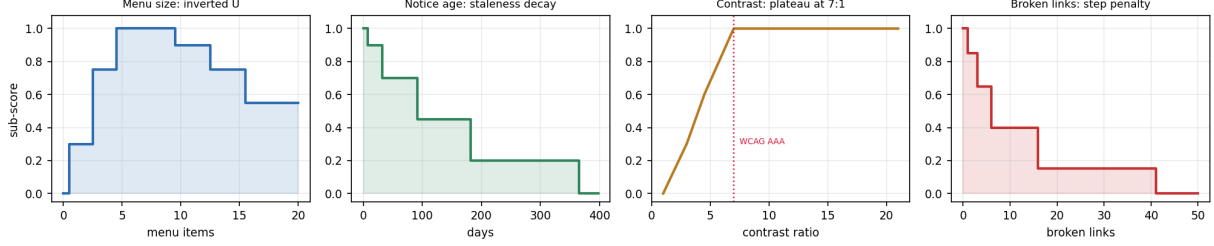


Figure 1: The four non-linear response curves used inside the dimension sub-scores.

- **Menu size — an inverted U.** A menu with 0–2 items leads nowhere; 5–9 items indicates an organised site; 20 items is a wall of links. Neither “more is better” nor “less is better” describes this, so no monotone function can represent it.
- **Notice age — staleness decay.** A post from last week and one from last year are not 51 weeks apart in value. “Recent” and “abandoned” are plateaus with a cliff between them.
- **Contrast — a plateau at 7:1.** WCAG AAA is 7:1. Beyond it, more contrast is not a better reading experience, so the curve stops rewarding rather than rising further.
- **Broken links — a step penalty.** One dead link is an oversight; forty is neglect. The penalty is not proportional to the count.

3.4 The resulting target

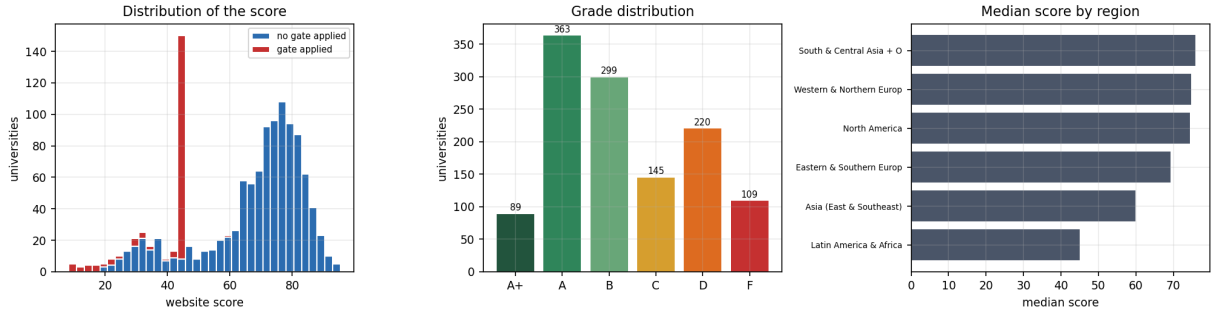


Figure 2: The website score across 1,225 universities. The shoulder near 45 in the left panel is the population to which a gating rule applies.

Mean	64.4	Grade A+ (≥ 85)	89
Standard deviation	18.4	Grade A (75–85)	363
Minimum	8.3	Grade B (65–75)	299
Maximum	95.4	Grade C (50–65)	145
Universities gated	186	Grade D (35–50)	220
		Grade F (< 35)	109

4 Experimental Setup

4.1 Train / test split

The dataset is split **80% training (979 universities)** / **20% testing (246 universities)**, stratified on score band so that both halves span the full quality range, with a fixed random seed so the split is reproducible. Stratification holds to within 1.5 percentage points in every band.

The test set is used once, in Section 6. No algorithm, hyper-parameter or threshold was selected using it. Model selection and tuning use 5-fold cross-validation on the training set only.

4.2 Algorithms compared

Twelve algorithms spanning six families, all trained on identical data. Feature scaling is applied inside a pipeline for the distance-, kernel- and gradient-based methods that require it, and deliberately not for tree methods, which are invariant to monotone rescaling.

Family	Algorithms
Baseline	Mean predictor
Linear	Linear Regression, Ridge ($\alpha = 1$), Lasso ($\alpha = 0.1$)
Instance	k -Nearest Neighbours ($k = 5$)
Kernel	Support Vector Regression (RBF, $C = 100$)
Neural	Multi-Layer Perceptron (128×64 , $L_2 = 10^{-2}$)
Tree	Decision Tree
Ensemble	Random Forest, Extra Trees, Gradient Boosting, LightGBM

4.3 Evaluation metrics

R^2 (variance explained), MAE (mean absolute error, in score points, directly interpretable), RMSE (penalises large errors) and **Spearman** ρ , which measures rank agreement and is the metric that matters most because the deliverable is a league table.

5 Model Comparison

Table 3: Twelve algorithms on the held-out test set, ordered by R^2 . Cross-validation figures are 5-fold on the training set only.

	Model	Family	R^2	MAE	RMSE	Spearman	CV R^2
1	LightGBM	ensemble	0.992	1.26	1.71	0.992	0.986 ± 0.001
2	Gradient Boosting	ensemble	0.989	1.51	1.94	0.990	0.985 ± 0.003
3	Neural Net (MLP)	neural	0.985	1.72	2.27	0.984	0.943 ± 0.035
4	Random Forest	ensemble	0.971	2.35	3.22	0.969	0.962 ± 0.004
5	SVR (RBF kernel)	kernel	0.961	2.48	3.72	0.979	0.955 ± 0.016
6	Linear Regression	linear	0.959	2.56	3.80	0.984	0.948 ± 0.009
7	Ridge	linear	0.959	2.60	3.80	0.984	0.945 ± 0.012
8	Lasso	linear	0.956	2.82	3.93	0.983	0.946 ± 0.007
9	Extra Trees	ensemble	0.945	3.13	4.41	0.939	0.917 ± 0.010
10	Decision Tree	tree	0.915	3.78	5.47	0.926	0.896 ± 0.011
11	k -NN ($k = 5$)	instance	0.786	5.97	8.67	0.875	0.770 ± 0.025
12	Mean baseline	baseline	0.000	15.73	18.76	—	-0.010 ± 0.008

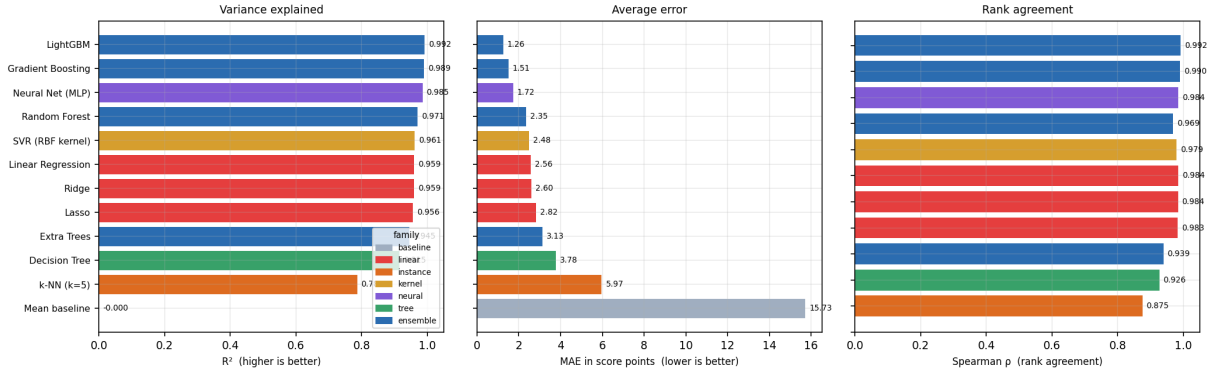


Figure 3: Model comparison on the held-out test set, coloured by algorithm family.

Three observations. First, every algorithm beats the mean baseline decisively, so the attributes carry real signal. Second, gradient-boosted ensembles lead on all four metrics. Third, k -NN is the clear worst: in a 71-dimensional, largely binary space, Euclidean distance is a poor measure of similarity — a textbook illustration of the curse of dimensionality.

5.1 Cross-validation

A single split can flatter or punish a model by chance, so the ordering was re-checked with 5-fold cross-validation on the training set alone.

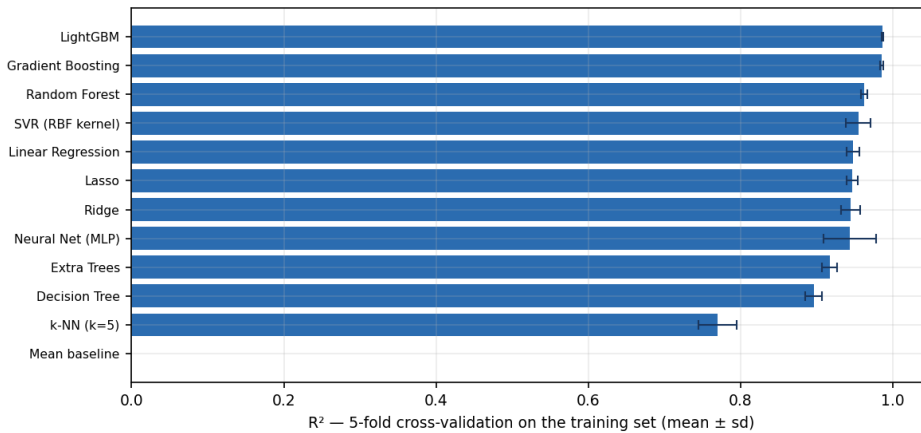


Figure 4: Cross-validated R^2 (mean $\pm$ standard deviation) on the training set.

The cross-validation ordering and the test-set ordering agree at Spearman $\rho = 0.888$, so the comparison is not an artefact of one particular split.

A practical note on the neural network

The MLP was the only model that required the *target* to be scaled as well as the inputs. Trained on the raw 0–100 scale, one fold in five diverged and cross-validated R^2 collapsed to -0.03 ± 1.88 . Wrapping it in a target-scaling transformer and adding L_2 regularisation stabilised it at 0.943 ± 0.035 . The tree-based models needed no equivalent care, which is part of why they are the practical choice at this dataset size.

6 Hyper-parameter Tuning and Final Evaluation

Grid search over 81 combinations, 5-fold cross-validated on the training set:

Parameter	Values searched (selected in bold)
n_estimators	300, 600, 900
learning_rate	0.03, 0.05 , 0.1
num_leaves	15 , 31, 63
min_child_samples	5 , 10, 20
Best cross-validated R^2 : 0.9892	

Table 4: Final model on the held-out test set — 246 universities, evaluated once.

R^2	0.9931
MAE (score points)	1.11
RMSE	1.56
Spearman ρ	0.9931
Kendall τ	0.9374
Predictions within ± 1 point	57.7%
Predictions within ± 2 points	83.3%
Predictions within ± 5 points	99.2%

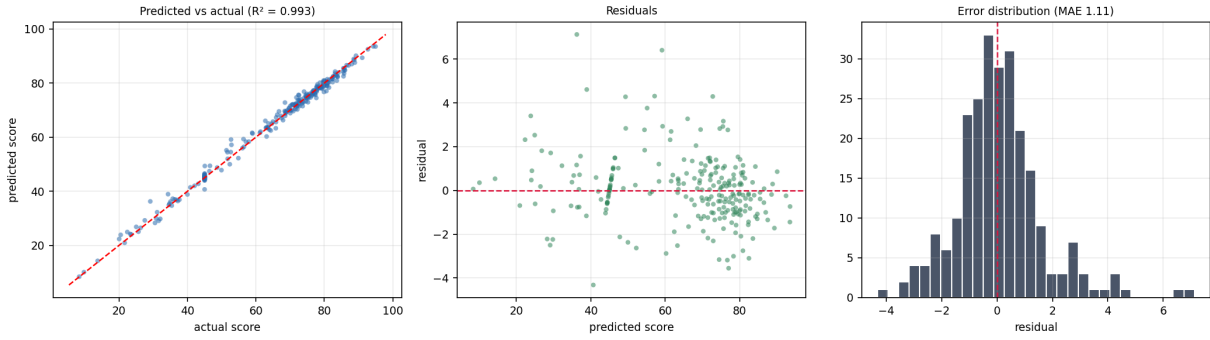


Figure 5: Held-out predictions. Residuals show no systematic pattern with predicted value.

7 Why the Algorithms Differ

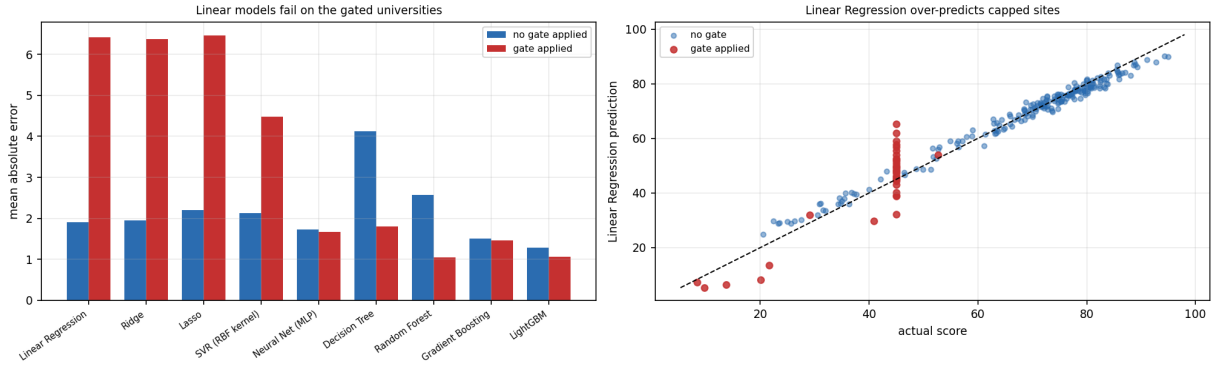
All twelve algorithms saw identical features and identical training data, yet R^2 ranged from 0.786 to 0.992. Attributing that spread to “model capacity” explains nothing. The real cause is specific, and it can be demonstrated.

The scoring model contains **gates** — hard caps at 45 and 60 points. A linear model computes $\sum_j \beta_j x_j$ and has no way to express “*whatever else is true, stop here*”. A decision tree does: a split is precisely that statement.

If this is the mechanism, then linear models should fail specifically on gated universities and behave normally elsewhere. Splitting the test error accordingly:

Table 5: Mean absolute error on the 36 gated versus 210 ungated test universities.

Model	Family	MAE gated	MAE ungated	ratio
Linear Regression	linear	6.41	1.91	3.37×
Ridge	linear	6.37	1.96	3.25×
Lasso	linear	6.47	2.20	2.94×
SVR (RBF)	kernel	4.49	2.14	2.10×
Neural Net (MLP)	neural	4.31	3.33	1.29×
Decision Tree	tree	1.80	4.12	0.44×
Random Forest	ensemble	1.05	2.57	0.41×
Gradient Boosting	ensemble	1.46	1.51	0.96×
LightGBM	ensemble	1.07	1.29	0.83×

**Figure 6:** Left: error split by whether a gate applied. Right: linear regression systematically over-predicts the capped universities (red), because it sums the content the site does have and never applies the cap.

The result worth taking away

Every linear model is roughly 3× worse on gated universities than on ungated ones. Every tree-based model is *as good or better* on them. The performance gap between algorithm families in Section 5 is therefore not a vague matter of capacity — it is the concrete inability of an additive model to represent a threshold, and it is visible on 36 specific universities.

8 What Drives a Website’s Score

Feature importance is computed by **gain** — how much each split reduced the loss — rather than by split count. The distinction matters: split-count importance flatters continuous attributes, which get split many times, over binary ones, which are split once. Under split-count importance, load time appeared to be the single most important feature; under gain it accounts for well under 1%.

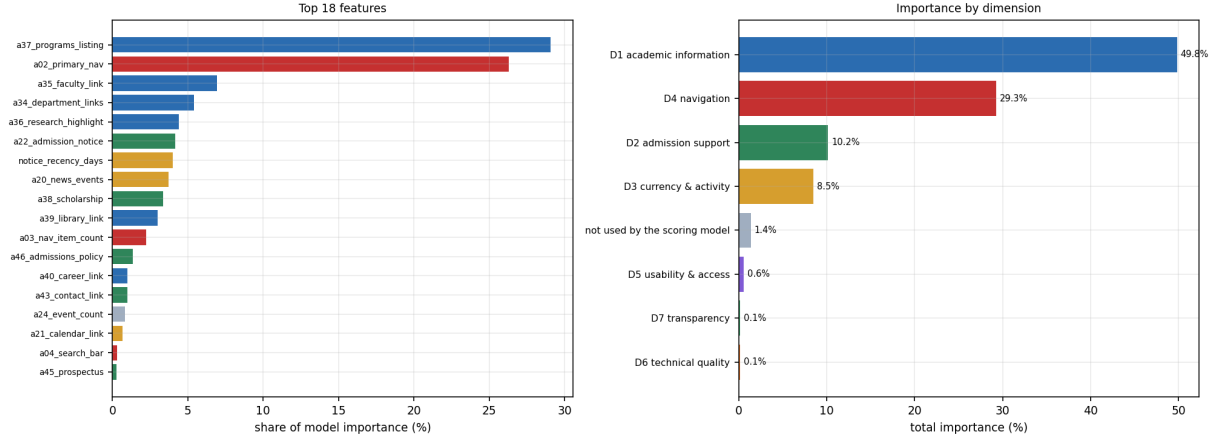


Figure 7: Gain-based feature importance for the tuned model, individually and aggregated by scoring dimension.

Table 6: Declared dimension weight against realised model importance.

Dimension	Declared	Realised	Interpretation
D_1 Academic information	28	49.8	the largest single block of variation between sites
D_4 Navigation	15	29.3	inflated because the navigation gate lives here
D_2 Admission support	22	10.2	
D_3 Currency, activity	15	8.5	
D_5 Usability, access	10	0.6	near-constant across sites, so it separates almost nobody
D_6 Technical quality	7	0.1	98.9% of sites use HTTPS; there is little left to explain
D_7 Transparency	3	0.1	

Two conclusions follow. First, **a declared weight is not the same as realised influence**: a dimension can only differentiate universities to the extent that they actually differ on it, and D_5 – D_7 are close to universal. This mirrors a known result in the website-evaluation literature, where a scheme allocating 40% of its score to performance was later found to drive under 3% of its ranking variance.

Second, the model **independently recovered the gating rule**. The single attribute `a02_primary_nav` accounts for **26.3% of all gain** — far beyond its arithmetic contribution of 3.75 points inside D_4 — because the model learned that this one binary determines whether a 45-point ceiling applies. This is a satisfying confirmation that the mechanism identified in Section 7 is what the model actually uses.

9 Rankings

Global top 10			Bangladesh top 8 (of 22)			
#	University	Score	#	University	Global	Score
1	University of New England (AU)	95.4	1	North South University	45	87.4
2	University of Porto (PT)	95.0	2	RUET	79	85.5
3	Eotvos Lorand University (HU)	95.0	3	CUET	114	83.8
4	University of Strasbourg (FR)	94.3	4	KUET	115	83.8
5	Southern Cross University (AU)	93.3	5	AIUB	136	83.1
6	City, University of London (UK)	92.9	6	BRAC University	145	82.7
7	Gauhati University (IN)	92.7	7	Univ. of Professionals	153	82.5
8	Nanyang Technological Univ. (SG)	92.7	8	Patuakhali S&T Univ.	235	80.2
9	RWTH Aachen University (DE)	92.6				
10	Sophia University (JP)	92.4				

Country rankings are produced for the **18 countries** with at least 20 universities in the dataset, covering 779 universities. For the 169 rows whose **country** field holds a regional label rather than a country, no country ranking is produced, because it would not be one.

10 Limitations

These bound how the ranking should be read and are stated plainly.

1. **The dataset cannot see visual design.** Every attribute is a presence flag, a count or a measurement. Whether a page is cluttered, dated-looking or simply unpleasant to use is not among the 71 columns and therefore cannot be in the score. A site that satisfies every content requirement will score well even if a human would find it ugly. Closing this gap would require screenshots and human raters, not further modelling.
2. **Approximately 2% of rows are extraction failures.** A small number of universities returned almost nothing to the crawler — no navigation, no programmes, no contact — which is characteristic of JavaScript-rendered pages the extractor could not read rather than of genuinely empty websites. These fall to the bottom of the ranking. The top of the ranking is unaffected, but the bottom should not be quoted without this caveat.
3. **The score reflects the landing page,** not the entire site and not the institution. An excellent admissions portal that is not linked from the front page is invisible to this method.
4. **The scoring weights are a documented judgement, not a measured truth.** They were fixed before scoring and are stated in full, but a different reasonable analyst would choose somewhat different weights and obtain a somewhat different ranking. The sensitivity of the ranking to the weights was not measured.
5. **Region and data collector are perfectly confounded.** Each of six collectors covered exactly one region, so regional differences in a measurement such as load time cannot be separated from differences in how it was measured. Load time is region-standardised for this reason, but the confound cannot be removed.
6. **The crawl is a single snapshot.** Websites change continuously; the freshness attributes in particular would differ on another day.

11 Conclusion

This project set out to automate the assessment of university website quality. We defined a seven-dimension scoring model grounded in what a prospective student needs, applied it to 1,225 universities, and trained twelve regression algorithms to predict the resulting score from raw

landing-page attributes. The tuned LightGBM model reaches $R^2 = 0.993$ with a mean absolute error of 1.11 points on a test set held out from every fitting and tuning decision, and orders the universities at Spearman $\rho = 0.993$.

The comparative study is the more instructive half. Performance across the twelve algorithms was not a smooth gradient of capability but a split along one specific structural line: models that can represent a threshold and models that cannot. Isolating the 36 gated universities in the test set made the mechanism visible — linear models are $3\times$ worse on exactly those cases and normal elsewhere — and gain-based importance analysis confirmed that the winning model had independently identified the gating attribute, assigning it 26% of all loss reduction.

Two methodological points generalise beyond this dataset. Missing values must be filled according to what absence *means*, not by a default statistic: median-filling one attribute here would have described 468 abandoned notice boards as updated yesterday. And a declared weight is not a realised influence: three dimensions carrying 20 points of the scoring model account for under 1% of what the trained model uses, simply because almost every university scores alike on them.

Future work

Adding screenshot-derived visual features and a small panel of human ratings would address the principal limitation. Replacing the extractor with a headless browser would eliminate the JavaScript-rendering failures. Finally, a sensitivity analysis over the dimension weights would quantify how much of the ranking is a property of the websites and how much is a property of our scoring choices.

A Repository contents

Path	Contents
data/university_website_scores.csv	all 1,225 universities, scored and ranked
data/train.csv, data/test.csv	the 979 / 246 split (80 / 20, seed 42)
data/data_dictionary.csv	every column, its dimension and its range
data/weka/	the same splits in ARFF format
notebook/University_Website_Quality.ipynb	the full analysis, executed
src/build_dataset.py	raw file $\rightarrow$ scored dataset $\rightarrow$ splits
src/make_figures.py	regenerates every figure in this report
model/final_model.joblib	the tuned model
results/	model comparison, importance, metrics as CSV

Environment. Python 3.13, pandas 2.3.3, numpy 2.4.4, scikit-learn 1.8.0, LightGBM 4.6.0. All random seeds fixed; the pipeline is reproducible end to end by running `src/build_dataset.py` followed by the notebook.